Part 2: Candidate Mechanism Local Projections

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1 Overview

This section estimates local projections to examine candidate transmission mechanisms of geopolitical risk (GPR) shocks. The analysis uses quarterly data from 1986Q1 to 2025Q4, with 160 observations in the full database. The candidate outcomes include real oil prices, inflation expectations, real activity variables, and financial-market volatility.

The empirical exercise proceeds in two stages. First, I construct standardized innovations from the World, Threat, and Act GPR series using univariate AR(1) residuals. Second, I estimate level-change local projections for each outcome and each shock series.

The baseline specification sets the maximum horizon to $H = 12$ quarters and includes $L = 4$ lags of GPR levels and macro controls. The number of outcome lags is selected by BIC from $1, \ldots, 4$ at each horizon. As an extended-horizon robustness check, I also estimate a richer specification with $H = 36$, $L = 12$, and BIC-selected outcome lags from $1, \ldots, 12$.

2 Empirical Specification

2.1 Stage 1: AR(1) GPR Innovations

For each GPR series $s \in \{\text{World}, \text{Threat}, \text{Act}\}$, I estimate the following AR(1) process:

$$g_t^{(s)} = \mu^{(s)} + \phi^{(s)} g_{t-1}^{(s)} + u_t^{(s)}, \tag{1}$$

where $g_t^{(s)}$ denotes the log level of the corresponding GPR index.

The estimated residual is standardized as

$$z_t^{(s)} = \frac{\hat{u}_t^{(s)} - \bar{u}^{(s)}}{\widehat{\sigma}_{u^{(s)}}}. \tag{2}$$

Thus, $z_t^{(s)}$ is interpreted as a one-standard-deviation innovation to the corresponding GPR component.

In the current sample, the AR(1) regressions produce R^2 values of 0.483, 0.470, and 0.516 for the World, Threat, and Act GPR series, respectively. These values indicate that a nontrivial part of GPR variation is persistent, while the residual component captures the unexpected movement relative to the index's own immediate past.

2.2 Stage 2: Level-Change Local Projections

For each outcome y, shock series $s \in \{\text{World}, \text{Threat}, \text{Act}\}$, and horizon $h = 0, \ldots, H$, I estimate the following local projection separately:

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h z_t^{(s)} + \sum_{j=1}^{I_h} \rho_{h,j} y_{t-j} + \sum_{j=1}^{L} \gamma_{h,j} g_{t-j}^{(s)} + \sum_{j=1}^{L} \theta'_{h,j} C_{t-j} + \varepsilon_{h,t}. \tag{3}$$

For notational simplicity, the coefficients are written without outcome- and shock-specific superscripts, but they are estimated separately for each outcome-shock pair and each horizon. The dependent variable is $y_{t+h} - y_{t-1}$, so the coefficient β_h is already a level-change response from the period immediately before the shock to horizon h. Therefore, the impulse responses should not be cumulated across horizons.

The control vector is

$$C_t = (\text{Unemp}_t, \, \text{FFR}_t)'. \tag{4}$$

VIX is not included as a control. Instead, log(VIX) is treated as an outcome variable, allowing the financial-risk channel itself to be examined rather than partialled out.

In the baseline specification, $H = 12$, $L = 4$, and I_h is selected by BIC from $\{1, 2, 3, 4\}$ separately for each outcome, shock, and horizon. In the robustness specification, $H = 36$, $L = 12$, and I_h is selected by BIC from $\{1, \ldots, 12\}$.

3 Lag Selection

Table 1 summarizes the BIC-selected outcome lag patterns in the baseline specification. The lag structure is generally parsimonious. For inflation expectations and log(VIX), BIC selects one outcome lag almost uniformly. Oil prices select longer lags only at the first few horizons. Real activity variables, especially investment and GDP, select longer lags at some horizons, reflecting stronger persistence.

Table 1: Summary of BIC-Selected Outcome Lags in the Baseline Specification

Outcome block	BIC-selected lag pattern
Oil prices	WTI selects two lags at $h = 0$ and one lag afterward. Gasoline selects three or four lags at the first two horizons, then one lag afterward.
Inflation expectations	Michigan, current SPF, and two-step-ahead SPF mostly select one lag across all horizons and shocks.
Real activity	Consumption mostly selects one lag after $h = 0$. Investment selects four lags at short horizons and three or two lags at later horizons. GDP selects longer lags at some middle horizons.
Financial risk	log(VIX) selects one lag across all horizons and shocks.

This pattern suggests that the main results are not driven by excessive autoregressive controls. Rather, BIC selects short outcome dynamics for most variables, while allowing more persistent real activity variables to retain longer own-lag structures.

4 Baseline Results: $H = 12$, $L = 4$

Figures 1 and 2 report the baseline local projection results using $H = 12$ and $L = 4$. The number of outcome lags is selected by BIC from $1, \ldots, 4$ at each horizon. The left column shows the response to the World GPR shock, while the right column compares the responses to Threat and Act shocks. Shaded areas indicate 90 percent Newey–West HAC confidence intervals.

Figure 1 focuses on oil prices and inflation expectations. The oil-price responses show a relatively stronger short-run reaction to Threat shocks, especially for real WTI prices. Inflation expectation responses are smaller overall, with current SPF expectations responding more visibly than medium-term SPF expectations.

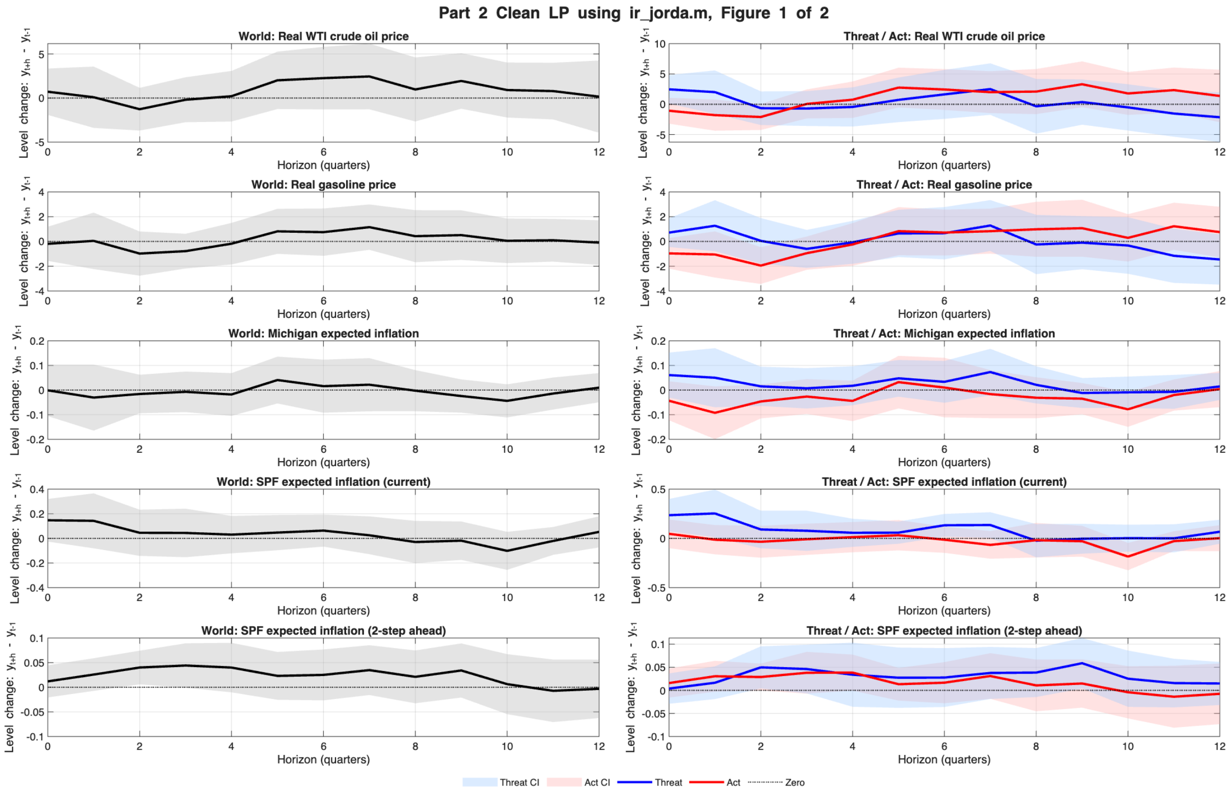

Figure 1: Baseline level local projections: Oil price and Inflation

Figure 2 reports the responses of real activity and financial-market volatility. Consumption and GDP responses are muted, while investment shows a positive response in the short run. The response of log(VIX) is small, suggesting that the financial-volatility channel is not dominant in the baseline specification.

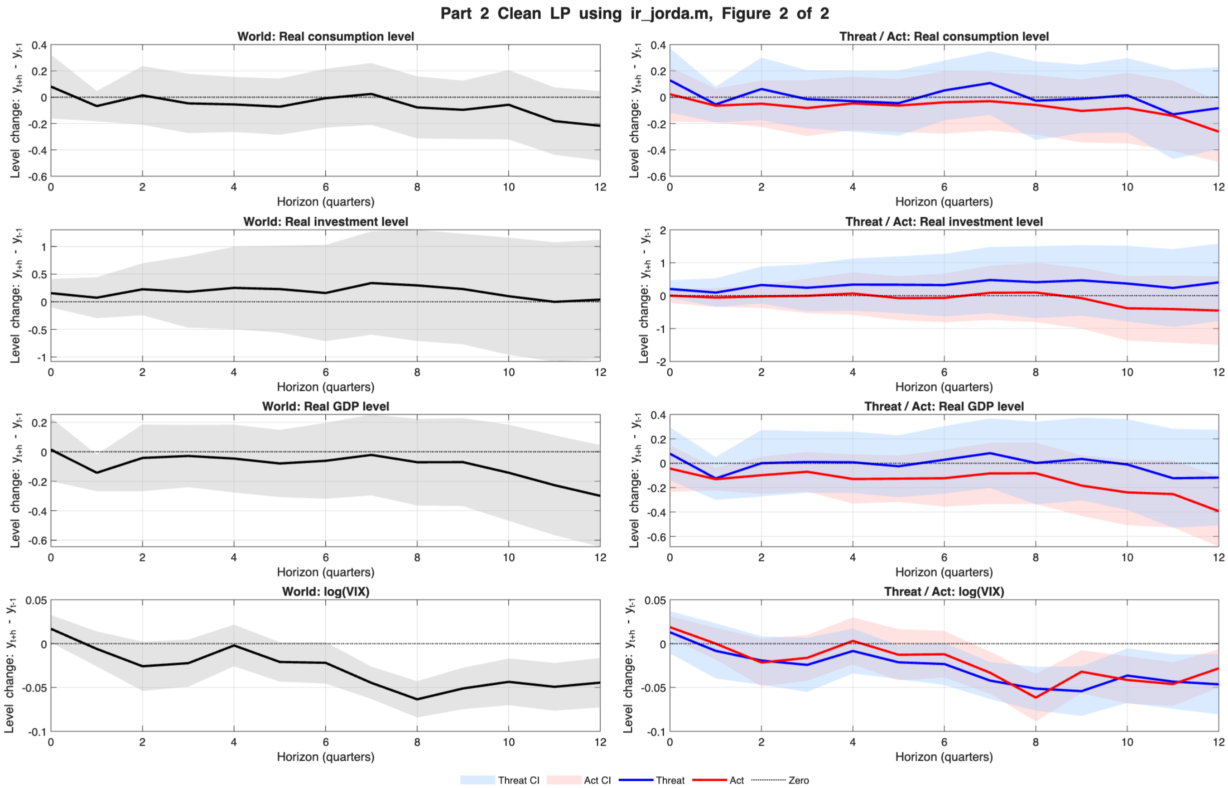

Figure 2: Baseline level local projections: Real activity and financial market volatility

4.1 Oil-Price Channel

The oil-price responses show the clearest short-run difference between Threat and Act shocks. A Threat shock raises real WTI prices on impact, while an Act shock initially moves WTI in the opposite direction. This pattern suggests that oil markets respond strongly to anticipatory geopolitical risk, while realized events may either have been partly priced in or may coincide with concerns about weaker demand.

Real gasoline prices display a weaker and more mixed response. The response is less pronounced than WTI, consistent with slower pass-through from crude oil prices to retail gasoline prices and with the influence of refining margins and domestic factors.

Table 2: Selected Oil-Price Responses

	World		Threat		Act	
Outcome	$h = 0$	$h = 4$	$h = 0$	$h = 4$	$h = 0$	$h = 4$
Real WTI	0.716	0.207	2.462	-0.437	-1.067	0.757
Real Gasoline	-0.188	-0.181	0.717	-0.071	-0.963	-0.241

4.2 Inflation Expectations Channel

Inflation expectation responses are relatively small. Current SPF inflation expectations respond more strongly to Threat shocks than to Act shocks, suggesting that professional forecasters revise short-run inflation expectations in response to anticipatory geopolitical risk. Michigan household inflation expectations show a weaker response.

The two-step-ahead SPF expectation remains close to zero across shocks and horizons. This suggests that medium-term inflation expectations remain relatively anchored, even when short-run expectations respond to geopolitical risk.

Table 3: Selected Inflation-Expectation Responses

	World		Threat		Act	
Outcome	$h = 0$	$h = 4$	$h = 0$	$h = 4$	$h = 0$	$h = 4$
Michigan Exp. Inflation	-0.001	-0.018	0.061	0.017	-0.044	-0.044
SPF Current Inflation	0.147	0.031	0.236	0.057	0.047	0.012
SPF Two-Step Ahead	0.012	0.040	0.004	0.034	0.016	0.039

4.3 Real-Activity Channel

Real activity responses are muted overall. Consumption and GDP responses are generally small and often close to zero. Investment is the main exception. The investment response is positive for World and Threat shocks at both $h = 0$ and $h = 4$, which is opposite to the standard real-options prediction that heightened uncertainty should delay investment.

This investment response should therefore be treated as an anomaly rather than as direct evidence against the uncertainty channel. Possible interpretations include the role of energy-related investment, the influence of geopolitical episodes associated with commodity-price booms, or the fact that lagged GPR-level controls absorb persistent geopolitical-risk dynamics differently from a specification using lagged shocks.

Table 4: Selected Real-Activity Responses

	World		Threat		Act	
Outcome	$h = 0$	$h = 4$	$h = 0$	$h = 4$	$h = 0$	$h = 4$
Consumption	0.081	-0.055	0.128	-0.030	0.022	-0.047
Investment	0.155	0.250	0.203	0.336	-0.001	0.063
Real GDP	0.015	-0.046	0.078	0.008	-0.044	-0.130

4.4 Financial-Risk Channel

The response of log(VIX) is small across all three shocks. The impact response is positive but modest, and the response generally decays over subsequent horizons. This suggests that, at the quarterly frequency, the equity-market volatility channel does not dominate the transmission of GPR shocks in this specification.

Table 5: Selected Financial-Risk Responses

	World		Threat		Act	
Outcome	$h = 0$	$h = 4$	$h = 0$	$h = 4$	$h = 0$	$h = 4$
log(VIX)	0.017	-0.002	0.013	-0.008	0.019	0.003

5 Robustness Check: $H = 36$, $L = 12$

As a robustness check, I extend the horizon to $H = 36$ quarters and increase the lag length for GPR levels and macro controls to $L = 12$. The outcome lag length is again selected by BIC, now from $1, \ldots, 12$. This specification allows the responses to be traced over a longer horizon, but it is also more demanding because the effective sample size declines as h increases and the number of potential lags is larger.

The robustness exercise is therefore not used as the main specification. Instead, it is used to examine whether the qualitative patterns in the baseline specification remain visible when the horizon and lag structure are expanded.

Figure 3 reports the extended-horizon responses of oil prices and inflation expectations. Compared with the baseline results, the responses become more volatile at longer horizons, especially for oil prices. The short-run distinction between Threat and Act shocks remains visible, but the longer-horizon dynamics should be interpreted cautiously because the estimates are based on fewer effective observations.

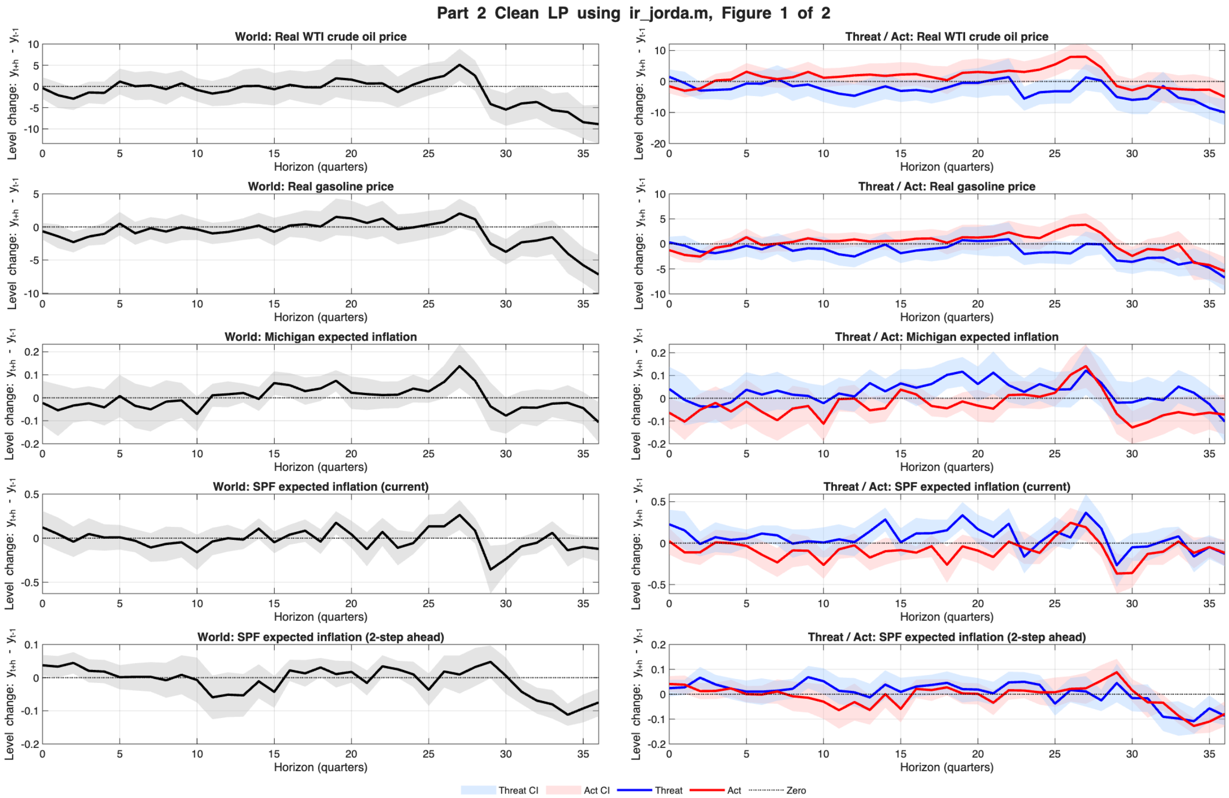

Figure 3: Robustness check of Figure 1

Figure 4 reports the extended-horizon responses of real activity and financial-market volatility. The longer-horizon specification reveals more persistent movements in investment, GDP, and log(VIX), but the estimates are also noisier than in the baseline. These patterns are useful for assessing robustness, but the baseline $H = 12, L = 4$ results remain preferable for the main interpretation.

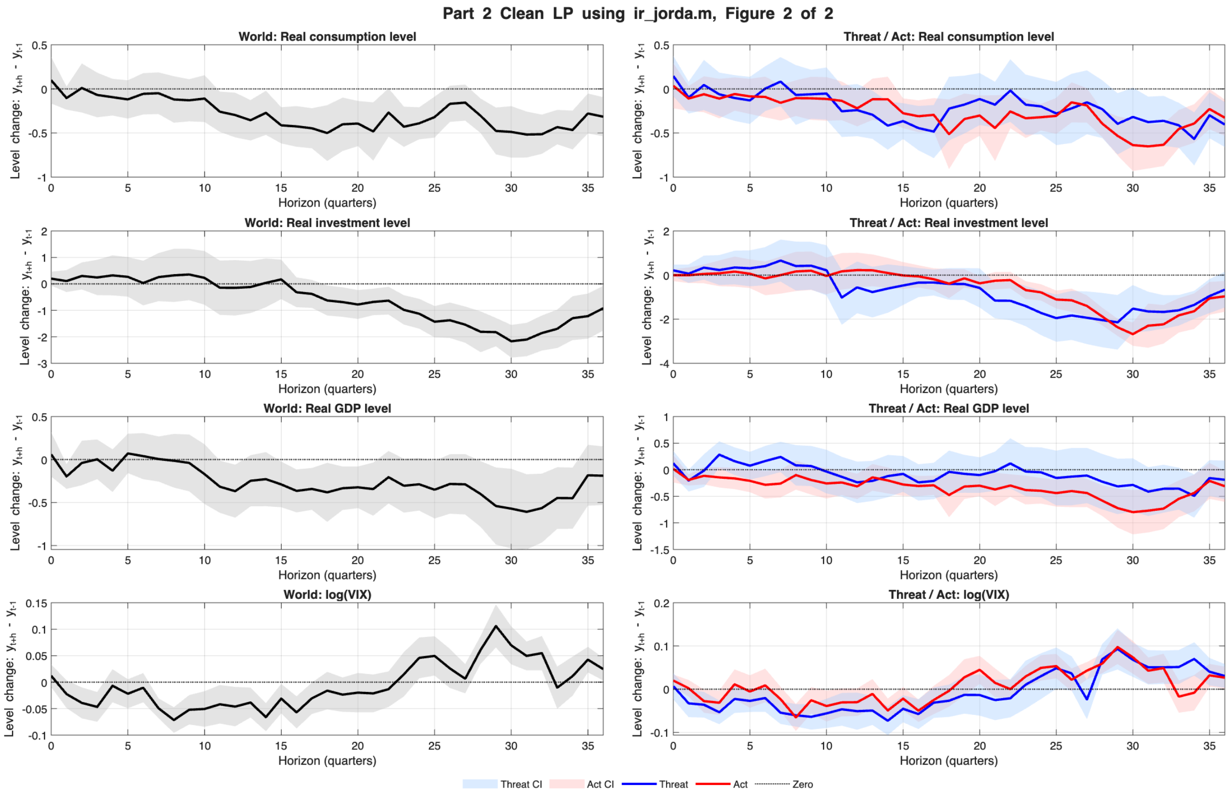

Figure 4: Robustness check of Figure 2

Overall, the baseline and robustness specifications deliver broadly similar qualitative messages. Threat shocks remain important for short-run commodity-price and inflation-expectation responses, medium-term inflation expectations remain relatively anchored, and the financial-volatility response is not dominant in the baseline. The extended-horizon results mainly show that long-run responses are more volatile and should be interpreted as robustness evidence rather than as the central set of results.

6 Main Takeaways

Several patterns emerge from the baseline specification.

First, Threat shocks generate the strongest short-run responses in oil prices and current SPF inflation expectations. This suggests that anticipatory geopolitical risk, rather than realized geopolitical acts alone, is important for commodity prices and short-run inflation expectations.

Second, medium-term inflation expectations remain relatively anchored. The two-step-ahead SPF response is small across World, Threat, and Act shocks, suggesting that geopolitical risk shocks do not generate large destabilizing movements in medium-term expected inflation.

Third, real activity responses are generally muted. Consumption and GDP responses are small, while investment displays a positive response. This investment response is anomalous relative to the standard uncertainty-channel prediction and should be examined further in robustness checks.

Fourth, the VIX response is small. This suggests that, in this quarterly specification, the financial-market volatility channel is not the dominant mechanism. Instead, the evidence points more strongly toward commodity-price and short-run expectation responses.

7 Caveats and Next Steps

The results should be interpreted as descriptive responses rather than structural causal effects. The AR(1) shock removes only the GPR index's own one-quarter persistence and may still be correlated with lagged oil-market or macroeconomic variables. The splitwise design also estimates World, Threat, and Act shocks separately, which avoids multicollinearity but does not test whether Threat or Act contains incremental explanatory power conditional on the others.

The next step is to compare these baseline results with alternative specifications. In particular, it would be useful to examine whether the main patterns are robust to an augmented shock construction that orthogonalizes GPR innovations with respect to additional oil-market and macroeconomic predictors. A joint specification including Threat and Act shocks together would also help test whether the two components carry distinct information.

8 Summary

Overall, the updated baseline specification with $H = 12$, $L = 4$, and BIC-selected outcome lags preserves the main qualitative conclusions. Threat shocks are associated with stronger short-run commodity-price and SPF inflation-expectation responses; medium-term inflation expectations remain anchored; VIX responses are small; and investment continues to display a positive response that is not easily reconciled with the standard uncertainty-channel mechanism.

The extended-horizon robustness check with $H = 36$ and $L = 12$ shows that the broad patterns remain visible, but the long-horizon responses are noisier and should be interpreted with caution.